

Cryptography - 30_09

✓ Theorem

Any family $\mathcal{H}$ of pairwise independent hash functions directly gives a $\mathcal{E} = \frac{1}{|\mathcal{T}|}$ -statistical secure MAC.

📖 Proof

Fix any $m \in \mathcal{M}, \tau \in \mathcal{T}$:

$$\Pr[\text{Tag}(K, m) = \tau] = \Pr[h(K, m) = \tau] = \frac{1}{|\mathcal{T}|}$$

by pairwise independent.

Similarly, for any m, m' such that $m \neq m'$ and $\forall \tau, \tau' \in \mathcal{T}$.

$$\Pr[\text{Tag}(K, m) = \tau \wedge \text{Tag}(K, m') = \tau'] = \Pr[h(K, m) = \tau \wedge h(K, m') = \tau'] = \frac{1}{|\mathcal{T}|^2}$$

also by pairwise independent

By applying Bayes:

$$\begin{aligned}\Pr[\text{Tag}(K, m') = \tau' | \text{Tag}(K, m) = \tau] &= \frac{\Pr[h(K, m') = \tau' \wedge h(K, m) = \tau]}{\Pr[h(K, m) = \tau]} \\ &= \frac{\frac{1}{|\mathcal{T}|^2}}{\frac{1}{|\mathcal{T}|}} = \frac{1}{|\mathcal{T}|}\end{aligned}$$

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Now we instantiate it (*we see how to build a pairwise hash function*).

Here is a construction:

Let p a prime:

$$h_{a,b}(m) = am + b \pmod{p}$$

$$k = (a, b) \in \mathbb{Z}_p^2$$

The key is two elements of $\mathbb{Z}_p$.

We also have that $\mathbb{Z}_p^2 = \mathcal{K}$ and $\mathbb{Z}_p = \mathcal{M} = \mathcal{T}$.

🔗 Lemma

The above $\mathcal{H}$ is pairwise independent.

🕒 Proof

For all $m, m' \in \mathbb{Z}_p$, with $m \neq m'$, and $z, z' \in \mathbb{Z}_p$

$$\Pr_{(a,b) \in \mathbb{Z}_p^2} [h_{a,b}(m) = \tau \wedge h_{a,b}(m') = \tau']$$

"sostituiamo le definizioni in un sistema di equazioni lineari, riscritto sotto forma di matrici"

$$\Pr_{(a,b) \in \mathbb{Z}_p^2} \left[\begin{cases} am + b = \tau \pmod{p} \\ am' + b = \tau' \pmod{p} \end{cases} \right]$$

$$\Pr_{(a,b) \in \mathbb{Z}_p^2} \left[\begin{pmatrix} m & 1 \\ m' & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \tau \\ \tau' \end{pmatrix} \right]$$

$$\Pr_{(a,b) \in \mathbb{Z}_p^2} = \left[\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} m & 1 \\ m' & 1 \end{pmatrix}^{-1} \cdot \begin{pmatrix} \tau \\ \tau' \end{pmatrix} \right]$$

We know that the determinant of the m, m' matrix is $\neq 0$, thus it's diagonally revertible. This because the determinant is $m - m'$, and because $m \neq m'$ then it's $\neq 0$. So $\exists$ only one solution for $\begin{pmatrix} a \\ b \end{pmatrix}$.

We then have that the number of possible cases is 1, while the amount of possible keys is p^2 :

$$= \frac{1}{p^2}$$

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🔗 Exercise

Show that the above schema would not be 2-time statistically secure.

🔗 Exercise 2

Construct a 3-wise independent hash family (triplets of elements, distribution is uniform for 3 elements (check how the key grows with the message)).

Randomness extraction

Alice and Bob need a random key. How do they generate it?

As we will see later, randomness will be crucial for secure crypto.

There are two components in any randomness generator (for example Fortuna, the randomness generator for windows, but also `/dev/rand` for Linux):

- Randomness extraction: By measuring physical quantities we can get an unpredictable sequence of bits.
→ (It's not necessarily uniform, and expensive to generate.)
From this, extract random y (output of randomness extraction) which is short (like 256 bits).
- Expand it then to any amount (polynomial) using a pseudorandom generator (PRG), which requires computational assumptions.

We want to understand: how to extract from unpredictable source X (it's not uniform, we want to get something uniform out of it).

An example is the Von Neumann extractor. Assume we have $B \in \{0, 1\}$ such that $Pr[B = 0] = p < \frac{1}{2}$ (it's not uniform). He samples $b_1 \in B, b_2 \in B$. If it happens that $b_1 = b_2$ we resample, else we get an output with:

$$\begin{cases} 1 & \text{if } b_1 = 0, b_2 = 1 \\ 0 & \text{if } b_1 = 1, b_2 = 0 \end{cases}$$

Assuming it outputs something, this will be such that:

$$Pr[\text{Output } 0] = Pr[\text{Output } 1] = p \cdot (1 - p)$$

So we have that having a certain output is dependent on the fact that one value is 0 (probability of p), and the other is 1 ($1 - p$).

$$Pr[\text{No output after } n \text{ tries}] = (2 \cdot p \cdot (1 - p))^n$$

which becomes small for large enough n .

We want now to generalize this question. Our dream would be to design Ext (deterministic) taking random variable X and outputting uniform $Ext(X)$.

This is actually impossible. The source must be unpredictable.

✓ Definition (min-entropy)

The min-entropy of X is $H_\infty(X) = -\log(\max_x \{Pr[X = x]\})$.

≡ Example

Let $X \equiv Un$ uniform over $\{0, 1\}^n$.

We have that $H_\infty(X) = -\log\left(\frac{1}{2^n}\right) = n$.

How many bits would be unpredictable? n .

≡ Example 2

Let X be constant (a constant value). What's the max probability? It's 1. So $H_\infty = 0$.

What is the next best thing: Design Ext that extracts uniform $Y = Ext(X)$ for every X such that $H_\infty \geq k$. It's also impossible, even if $Ext(X) = b \in \{0, 1\}$ and $k = n - 1$ and $x \in \{0, 1\}^n$.

Here is why: Fix any $Ext : \{0, 1\}^n \rightarrow \{0, 1\}$. Let $b \in \{0, 1\}$ be the output maximizing $|Ext^{-1}(b)|$.



We can sure say that $|Ext^{-1}(b)| \geq 2^{n-1} = \frac{2^n}{2}$.

The bad X : Define X to be uniform over $Ext^{-1}(b)$. Since it is uniform:

$H_\infty(X) \geq n - 1$. But $Ext(X) = b$ always, so it's a constant, and thus we cannot use X like that because it's not uniform.

Solution: Swap the quantifiers.

✓ Definition (Seeded Extractor)

A function $Ext : \{0, 1\}^d \times \{0, 1\}^n \rightarrow \{0, 1\}^l$ (with d as the seed, n is the source) is a $(k, \mathcal{E})$ -extractor if

$\forall X$ such that $H_\infty \geq k$:

$Ext(S, X) \equiv U_l$ (the output of the extractor is uniform)

$\implies (S, Ext(S, X)) \approx_{\mathcal{E}} (S, U_l)$ ($\equiv$ in this case means equality of the random variable)

for $S \equiv U_d$ (uniform over $\{0, 1\}^d$)

(note that $(S, U_l) \equiv U_{l+d}$).

What does it mean? There is a standard way to measure distance between distribution:

We say that $Z \approx_{\mathcal{E}} Z' \iff SD(Z, Z') \leq \mathcal{E}$, where SD (statistical distance) is

$SD(Z, Z') = \frac{1}{2} \sum_z |Pr[Z = z] - Pr[Z' = z]| \leq \mathcal{E}$ (with Z and Z' being two distributions).

This is equivalent: $\forall$ unbounded adversary A :

$$|Pr[A(z) = 1 : z \in Z] - Pr[A(z) = 1 : z \in Z']| \leq \mathcal{E}$$

✓ Theorem (Leftover Hash Lemma)

Let $\mathcal{H} = \{h_s : \{0, 1\}^n \rightarrow \{0, 1\}^l\}_{s \in \{0, 1\}^d}$ be a pairwise independent family.

Then, $Ext(s, x) = h(s, x)$ is a $(k, \mathcal{E})$ -extractor for $k \geq l + 2 \cdot \log(\frac{1}{\mathcal{E}})^{-2}$.

s is now public, before it was secret.

🌀 Lemma

Let Y be a random variable over $\mathcal{Y}$, such that

$Col(Y) = \sum_{y \in \mathcal{Y}} Pr[Y = y]^2 \leq \frac{1}{|\mathcal{Y}|} \cdot (1 + 4\mathcal{E}^2)$ (collision probability, we sample

twice and both times we get y)

Then, $SD(Y; U) \leq \mathcal{E}$